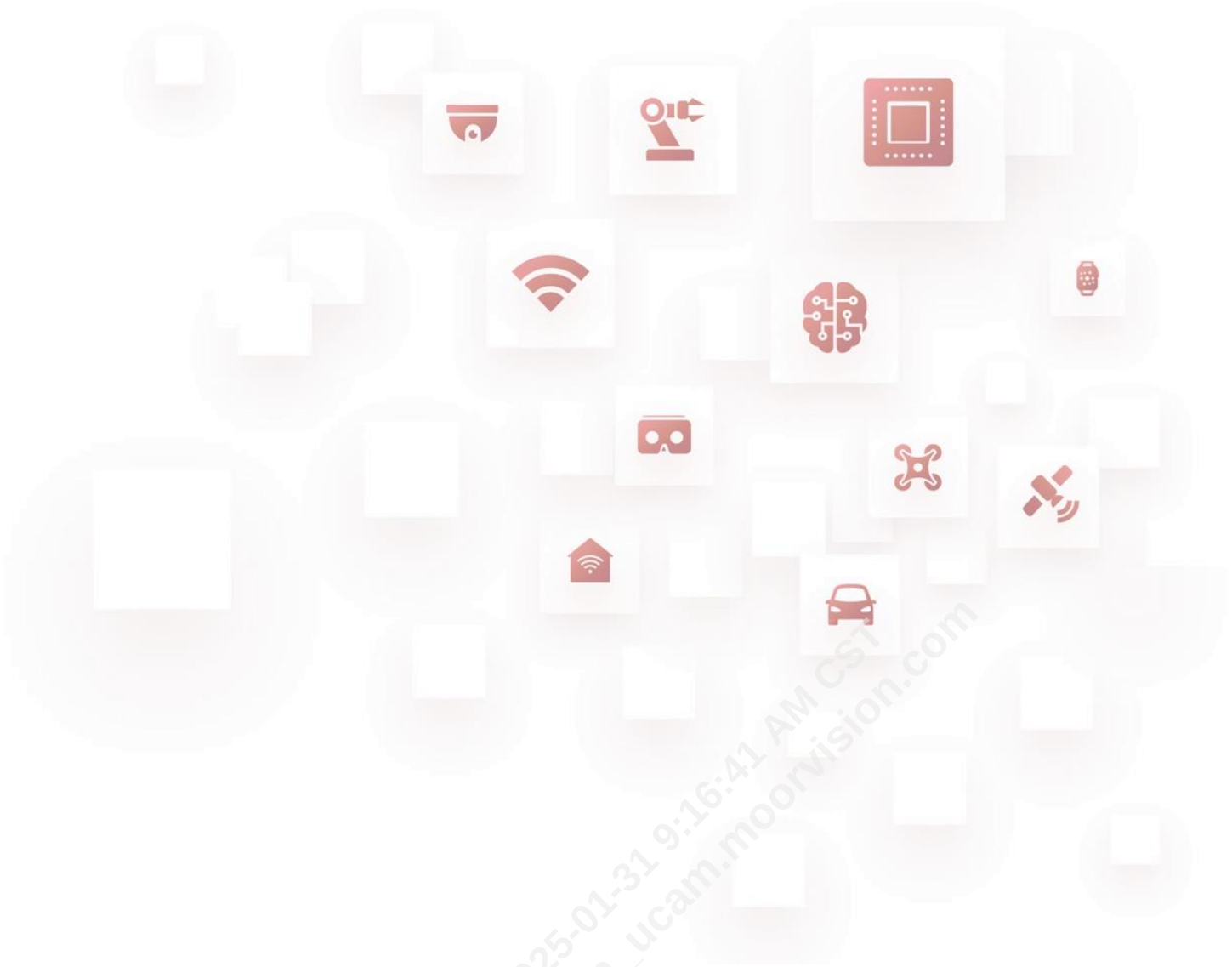
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Thundercomm TurboX C410/C610 Open Kit Linux Android

Software User Manual

Rev. V1.0.2
Jan 11, 2023

DN: tc-gla-2310.2

Revision History

Revision	Date	Description
1.0	Apr 01, 2021	Initial release.
1.0.1	Apr 01, 2022	• Update Figure 1-1. • Update Table 1-1. • Update Figure 1-2. • Add log output in Section 2.4. DisplayPort. • Remove *Section 2.12 Capture card with DMICs (digital microphones)* as this function is not suppported. • Fix format problems.
1.0.2	Jan 11, 2023	• Update Table 1-2. • Update 2.3. DSI LCD panel. • Add 2.14. HDMI IN. • Fix format problems.

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About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Product Overview

TurboX C410/C610 Open Kit is a cost-effective solution to evaluate the performance and build the prototype for smart cameras and intelligent IoT devices quickly. The Open Kit features rich interfaces, including USB Type A, Ethernet, and camera modules. It supports Android (Q4,2020) operating system, combining powerful artificial intelligence and machine learning functions to address increasing demand for smart cities, commerce, enterprise, homes, and vehicles applications.

➲ **NOTE:** "TurboX" referred to herein is the English text of our registered trademark TURBOX.

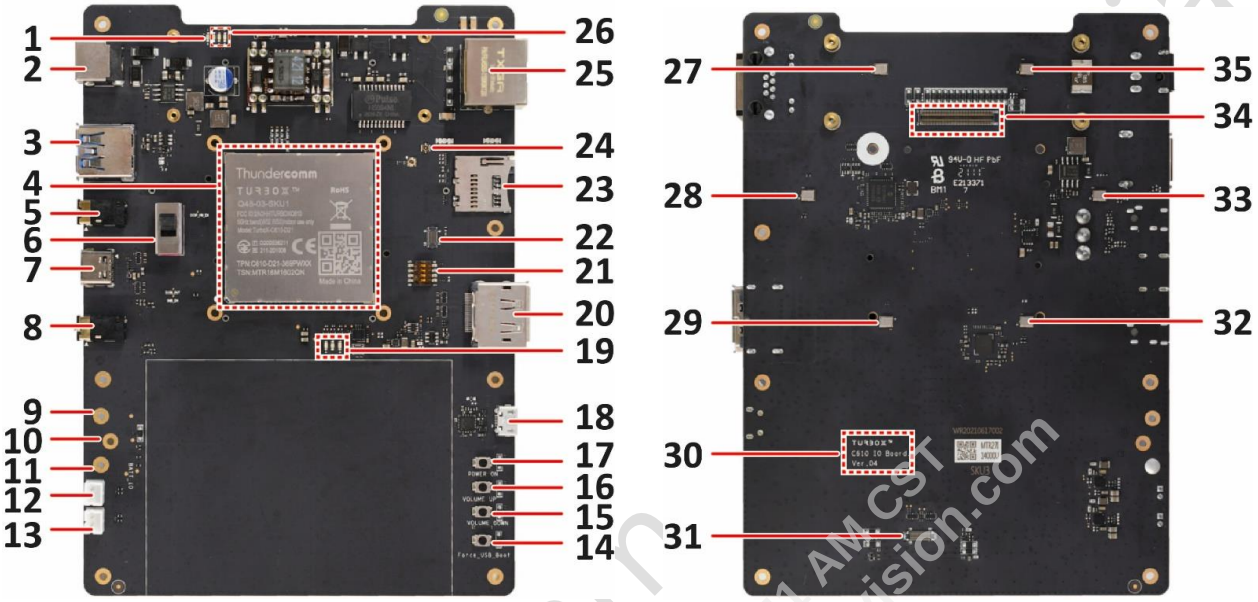


Figure 1-1. Board Layout (Hardware Version 04)

Table 1-1. Board layout

1. Power LED	13. Speaker connector 2	25. Ethernet connector
2. 12V DC in jack	14. Force USB Boot button	26. PMIC RGB LED
3. USB 3.1 Type-A connector (USB 2.0 protocol only)	15. Volume down button	27. Digital microphone 1
4. Heat sink fixing hole	16. Volume up button	28. Digital microphone 2
5. Line in connector	17. Power on button	29. Digital microphone 3
6. Power source selective switch	18. UART debug connector	30. Hardware version
7. USB 3.1 Gen1 Type-C connector	19. GPIO LED	31. LCD connector
8. Headset connector	20. DisplayPort	32. Digital microphone 4
9. VBAT + pad	21. Boot config switch	33. Digital microphone 5
10. BAT thermal pad	22. JTAG debug connector	34. Camera mezzanine board connector
11. VBAT - pad	23. Micro SD card slot	35. Digital microphone 6
12. Speaker connector 1	24. Wi-Fi/BT antenna connector	-

➲ **NOTE:** Camera Module and TFT-LCD are accessories to the Open Kit.

1.1. Features and specifications

Table 1-2. Features and specifications

Category	Description
System	TurboX C410/C610 SOM
Vertical Mezzanine Board (Camera board)	• Vertical A: Panorama - 2 x IMX577 • Vertical B: Conference: - 1 x IMX415 + 1 x HDMI IN
Display Interfaces	1 x DP; 1 x 5.0 TFT-LCD
Audio Interfaces	• 1 x 3.5mm headset • 6 x digital microphone • 2 x speaker out
General interfaces on mother board	• 1 x USB 3.1 Gen1 Type-C • 1 x USB 2.0 Type-A • 1 x Micro USB (for debug) • 1 x SD card • 1 x Gb Ethernet • 1 x Accelerometer + Gyro Sensor • 1 x Wi-Fi/BT antenna connector • 6 x LED • 4 x button (Volume up/down, Force USB Boot, Power on)
Power Supply	DC IN 12V, POE 15W, Battery Connector
Operating Environment	• Operation temperature: -25℃ ~ 75℃ • Operation humidity: 5%~95%, non-condensing
Dimension	120 x 250 mm
OS Support	Linux Android • LA1.0 (Android 10) • LA2.0 (Android 12)

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1.2. Product diagram

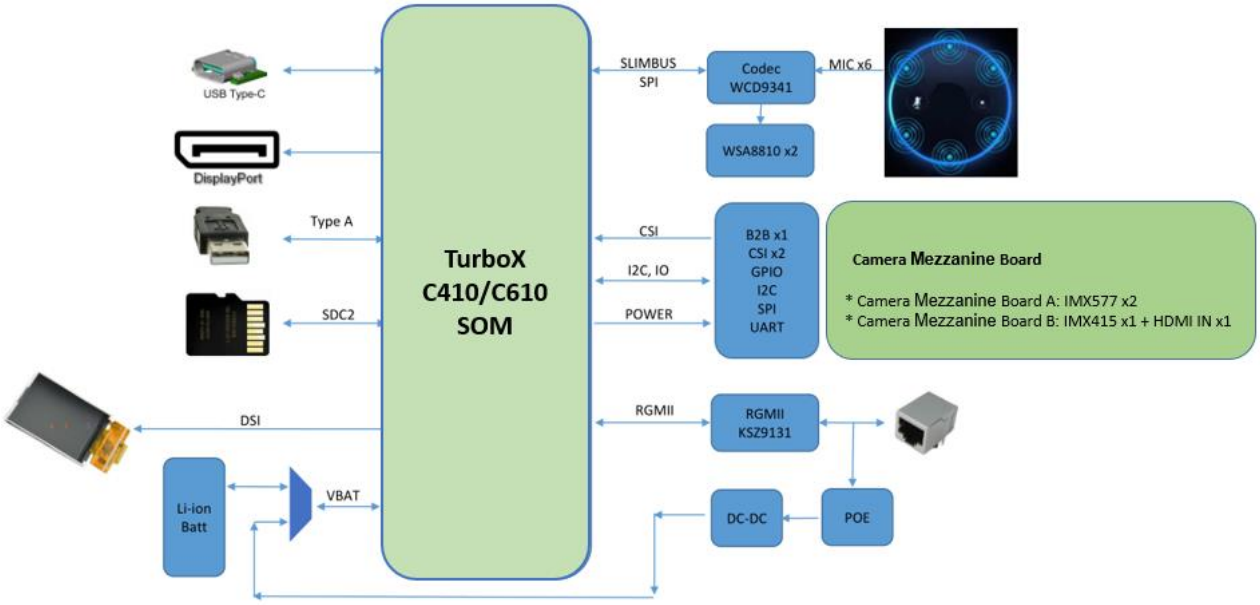


Figure 1-2.Block Diagram

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Chapter 2. Function Test

This chapter introduces the detailed testing steps of board function.

2.1. Debug port

Step 1.Connect the board to the computer via the **UART debug connector** (No. 18 in Figure 1-1).

Step 2.Use a UART tool you prefer, such as **Minicom**.

A new serial port will be added to your computer after the device is connected. Select the new serial port, and configure it to 115200 8N1. Refer to Figure 2-1 for the configuration of **Minicom 2.7**.



Figure 2-1.Minicom 2.7 Configuration Screen Capture

Step 3.Log in the board system with the following account credentials.

```
… …
[  OK  ] Started start dsp variants.
[  OK  ] Reached target Multi-User System.
         Starting Update UTMP about System Runlevel Changes...
[  OK  ] Started Update UTMP about System Runlevel Changes.

qsap 202001010057 qcs610-odk ttyMSM0
qcs610-odk login: root
Password: oelinux123
```

2.2. SD card slot

The SD card slot is hot-swappable. When inserted to the board, SD card shall automatically mount to */sdcard/* to access contents.

Run the following commands to test read and write speed of the SD card.

```
$ adb shell
# cd /mnt/media_rw/98C7-A27D
# dd if=/dev/urandom of=test.txt bs=30M count=2 conv=fsync
2+0 records in
2+0 records out
62914560 bytes (60 M) copied, 3.935880 s, 15 M/s
# echo 3 >/proc/sys/vm/drop_caches
# dd if=test.txt of=/dev/null
122880+0 records in
122880+0 records out
62914560 bytes (60 M) copied, 0.974147 s, 62 M/s
```

➲ **NOTE:** Replace `98C7-A27D` with your TF card directory name (each TF card corresponds to a directory).

2.3. DSI LCD panel

The board supports two LCD panel versions: ili9881c and ili9881p.

➲ **NOTE:**

- Software versions below LA1.0.Post-CS1.r002006 of LA1.0 baseline can only support ili9881c.
- LA1.0.Post-CS1.r002006 and later versions of LA1.0 baseline as well as all software versions of LA2.0 can support both ili9881c and ili9881p.

Step 1.Connect the LCD panel to the board via the **LCD connector** (No. 31 in Figure 1-1).



Figure 2-2.ili9881c LCD Panel



Figure 2-3.ili9881p LCD Panel

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Step 2. Download the corresponding CDT file of the desired LCD panel.

For detailed procedure, refer to *TurboX C410 C610 SOM_Flash CDT Image with TurboX Easy Flash User Guide*.

- **CDT file of ili9881c**: turbox-c610-cdt-opk_LCD_ili9881c.
- **CDT file of ili9881p:** turbox-c610-cdt-opk_LCD_ili9881p.

Step 3. Connect debug UART and press the **Power on button** (No. 17 in Figure 1-1) to boot up the system normally.

Refer to the corresponding log output.

```
Format: Log Type - Time(microsec) - Message - Optional Info
Log Type: B - Since Boot(Power On Reset),  D - Delta,  S - Statistic
S - QC_IMAGE_VERSION_STRING=BOOT.XF.3.1-00569-SM6150LZB-1
S - IMAGE_VARIANT_STRING=SDM6150LA
S - OEM_IMAGE_VERSION_STRING=8b8891cc45fb
…
S -    123623 - PBL, End
B -    143136 - SBL1, Start
…
UEFI Start     [ 1024] SEC
ASLR       : ON
DEP        : ON (RTB)
…
Loader Build Info: Oct 22 2020 19:44:02
VB: RWDeviceState: Succeed using devinfo!
Total DDR Size: 0x000000007ED00000
KeyPress:0, BootReason:0
Fastboot=0, Recovery:0
GetVmData: No Vm data present! Status = (0x3)
VM Hyp calls not present
Booting from slot (_a)
Booting Into Mission Mode
....
Update Device Tree total time: 77 ms
Start EBS       [13803]
BDS: LogFs sync skipped, Unsupported
App Log Flush : 6 ms
Exit EBS       [13823] UEFI End
[    0.000000] Booting Linux on physical CPU 0x0
[    0.000000] Linux version 4.14.117+ (daniel@TSBJ-FA-PC-03418) (clang version
8.0.12 for Android NDK) #17 SMP PREEMPT Wed Oct 21 16:03:34 CST 2020
[    0.000000] Boot CPU: AArch64 Processor [51df805e]
[    0.000000] Machine: IOT
[    0.000000] earlycon: msm_geni_serial0 at MMIO 0x0000000000880000 (options '')
```

Step 4. After image flash and system reboot, you will see the Android Home Screen via LCD panel in 3 minutes.

2.4. DisplayPort

Step 1.Connect the monitor (an accessory) to the board.

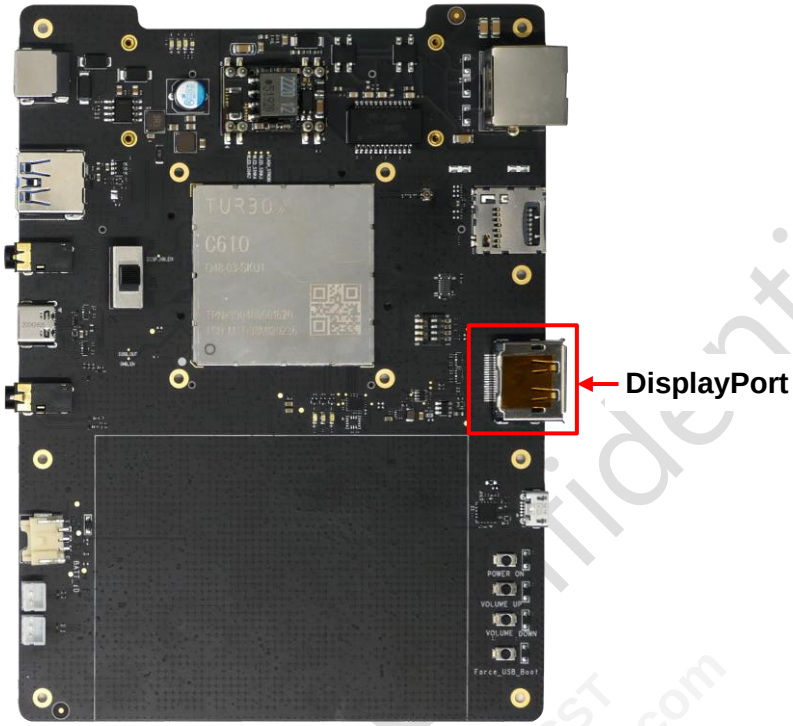


Figure 2-4.DisplayPort Location

Step 2.Connect debug UART and press the **Power on button** (No. 17 in Figure 1-1) to boot up the system normally.

Refer to the corresponding log output.

```
Format: Log Type - Time(microsec) - Message - Optional Info
Log Type: B - Since Boot(Power On Reset), D - Delta, S - Statistic
S - QC_IMAGE_VERSION_STRING=BOOT.XF.3.1.c3-00001-SM6150LZB-1
S - IMAGE_VARIANT_STRING=SDM6150LA
S - OEM_IMAGE_VERSION_STRING=b5765d9c8902
S - Boot Interface: eMMC
S - Secure Boot: Off
S - Boot Config @ 0x00786070 = 0x000000c1
S - JTAG ID @ 0x00786130 = 0x1011c0e1
S - OEM ID @ 0x00786138 = 0x00000000
S - Serial Number @ 0x00784138 = 0xc3a7ccb6
S - OEM Config Row 0 @ 0x00784188 = 0x0000000000000000
S - OEM Config Row 1 @ 0x00784190 = 0x0000000000000000
S - Feature Config Row 0 @ 0x007841a0 = 0x08d0200016388400
S - Feature Config Row 1 @ 0x007841a8 = 0xe014ffff00037fff
S - Core 0 Frequency, 1516 MHz
S - PBL Patch Ver: 3
S - PBL freq: 600 MHZ
D - 5633 - pbl_apps_init_timestamp
```

Step 3.After image flash and system reboot, you will see the Android Home Screen via the DisplayPort monitor in 3 minutes.

2.5. Touch panel

The touch panel starts to function once the LCD panel is installed successfully.

2.6. USB connector

2.6.1. Type-A USB connector

Step 1. Connect a mouse to the board via the **USB 3.1 Type-A connector** (No.3 in Figure 1-1) and check the information of the input device.

Step 2. Connect a USB mouse to the Type-A USB connector. For more information on the mouse information, input `# dmesg`.

```
# dmesg
[  153.084432] msm-dwc3 a800000.hsusb: DWC3 exited from low power mode
[  153.318298] usb 1-1: new low-speed USB device number 2 using xhci-hcd
[  153.471898] usb 1-1: New USB device found, idVendor=04ca, idProduct=0061
[  153.471903] usb 1-1: New USB device strings: Mfr=1, Product=2, SerialNumber=0
[  153.471906] usb 1-1: Product: USB Optical Mouse
[  153.471910] usb 1-1: Manufacturer: PixArt
[  153.489928] input: PixArt USB Optical Mouse as
/devices/platform/soc/a800000.hsusb/a800000.dwc3/xhci-hcd.0.auto/usb1/1-1/1-
1:1.0/0003:04CA:0061.0001/input/input5
[  153.548384] hid-generic 0003:04CA:0061.0001: input: USB HID v1.11 Mouse [PixArt
USB Optical Mouse] on usb-xhci-hcd.0.auto-1/input0
```

➲ **NOTE:** Constrained by memory configuration, the device does not support to automatically mount with a hard disk or USB device inserted. However, you can execute the mount commands below manually with a USB device or hard disk inserted.

```
$ adb root
$ adb shell
# cat /proc/partitions     //look hard/u disk imformation
You will see something like sda1
# mkdir storage/sdcard    //creat a mount point
# mount -t vfat /dev/block/sda1 /storage/sdcard    //mount your device
```

2.6.2. Type-C USB connector

Step 1.The board shall detect an external hub once connected by a Type-C USB OTG cable. Input `dmesg | grep usb` for detailed information.

```
# dmesg | grep usb
[   11.143236] usbpd usbpd0: Type-C Sink connected
[   11.169777] msm-usb-ssphy-qmp 88e6000.ssphy: USB QMP PHY: Update TYPEC CTRL(3)
[   11.182107] msm-dwc3 a600000.ssusb: DWC3 exited from low power mode
[   11.252470] usb usb3: New USB device found, idVendor=1d6b, idProduct=0002
[   11.259457] usb usb3: New USB device strings: Mfr=3, Product=2, SerialNumber=1
[   11.266872] usb usb3: Product: xHCI Host Controller
[   11.271896] usb usb3: Manufacturer: Linux 4.14.117+ xhci-hcd
[   11.277706] usb usb3: SerialNumber: xhci-hcd.1.auto
[   11.312942] usb usb4: New USB device found, idVendor=1d6b, idProduct=0003
[   11.319945] usb usb4: New USB device strings: Mfr=3, Product=2, SerialNumber=1
[   11.319948] usb usb4: Product: xHCI Host Controller
[   11.327482] usb usb4: Manufacturer: Linux 4.14.117+ xhci-hcd
[   11.339961] usb usb4: SerialNumber: xhci-hcd.1.auto
…
[   20.559551] f_cdev_alloc: port_name:at_usb0 (ffffffe79e160c80) portno:(0)
[   20.562356] f_cdev_alloc: port_name:at_usb1 (ffffffe781bcaa80) portno:(1)
[   20.564782] f_cdev_alloc: port_name:at_usb2 (ffffffe781bcda80) portno:(2)
[   24.877197] init: Service 'usbd' (pid 1137) exited with status 0
[   32.839059] msm-dwc3 a600000.ssusb: DWC3 exited from low power mode
[   32.877814] msm-dwc3 a800000.hsusb: DWC3 exited from low power mode
[   33.928630] msm-dwc3 a800000.hsusb: DWC3 in low power mode
[   33.979067] msm-dwc3 a600000.ssusb: DWC3 in low power mode
[   58.262364] msm-dwc3 a600000.ssusb: DWC3 exited from low power mode
[   58.508337] usb 3-1.4: new high-speed USB device number 3 using xhci-hcd
[   58.629668] usb 3-1.4: New USB device found, idVendor=05e3, idProduct=0608
[   58.629674] usb 3-1.4: New USB device strings: Mfr=0, Product=1, SerialNumber=0
[   58.629677] usb 3-1.4: Product: USB2.0 Hub
[   58.928293] usb 3-1.4.1: new high-speed USB device number 4 using xhci-hcd
[   59.041024] usb 3-1.4.1: New USB device found, idVendor=0b95, idProduct=772b
[   59.041030] usb 3-1.4.1: New USB device strings: Mfr=1, Product=2, SerialNumber=3
[   59.041034] usb 3-1.4.1: Product: AX88772C
[   59.041037] usb 3-1.4.1: Manufacturer: ASIX Elec. Corp.
[   59.041041] usb 3-1.4.1: SerialNumber: 000290
```

Step 2.Connect a USB storage device with OTG cable. Input `dmesg` for detailed information on the USB storage.

```
# dmesg
[   47.008360] usb 3-1.1: new high-speed USB device number 3 using xhci-hcd
[   47.142747] usb 3-1.1: New USB device found, idVendor=058f, idProduct=6387
[   47.142753] usb 3-1.1: New USB device strings: Mfr=1, Product=2, SerialNumber=3
[   47.142757] usb 3-1.1: Product: Mass Storage
[   47.142761] usb 3-1.1: Manufacturer: Generic
[   47.142764] usb 3-1.1: SerialNumber: A75BAA41
[   47.155700] usb-storage 3-1.1:1.0: USB Mass Storage device detected
[   47.159716] scsi host0: usb-storage 3-1.1:1.0
[   48.170623] scsi 0:0:0:0: Direct-Access     Generic  Flash Disk      8.07 PQ: 0
ANSI: 4
[   48.199335] sd 0:0:0:0: [sda] Write Protect is off
[   48.204322] sd 0:0:0:0: [sda] Mode Sense: 23 00 00 00
[   48.214369] sda: sda1
```

2.7. LED control

You can control the LED status with the following commands.

```
$ adb shell
# echo 0 > /sys/devices/platform/soc/soc:leds-gpios/leds/led_bt/brightness     //Turn off
# echo 1 > /sys/devices/platform/soc/soc:leds-gpios/leds/led_bt/brightness     //Turn on
# echo 0 > /sys/devices/platform/soc/soc:leds-gpios/leds/led_wifi/brightness   //Turn off
# echo 1 > /sys/devices/platform/soc/soc:leds-gpios/leds/led_wifi/brightness   //Turn on
# echo 0 > /sys/devices/platform/soc/soc:leds-gpios/leds/led_gen/brightness      //Turn off
# echo 1 > /sys/devices/platform/soc/soc:leds-gpios/leds/led_gen/brightness      //Turn on
# echo 255 > /sys/class/leds/red/brightness //Turn on
# echo 0 > /sys/class/leds/red/brightness //Turn off
# echo 255 > /sys/class/leds/green/brightness //Turn on
# echo 0 > /sys/class/leds/green/brightness //Turn off
# echo 255 > /sys/class/leds/blue/brightness //Turn on
# echo 0 > /sys/class/leds/blue/brightness //Turn off
```

2.8. Buttons

Refer to the figure below to locate the buttons.

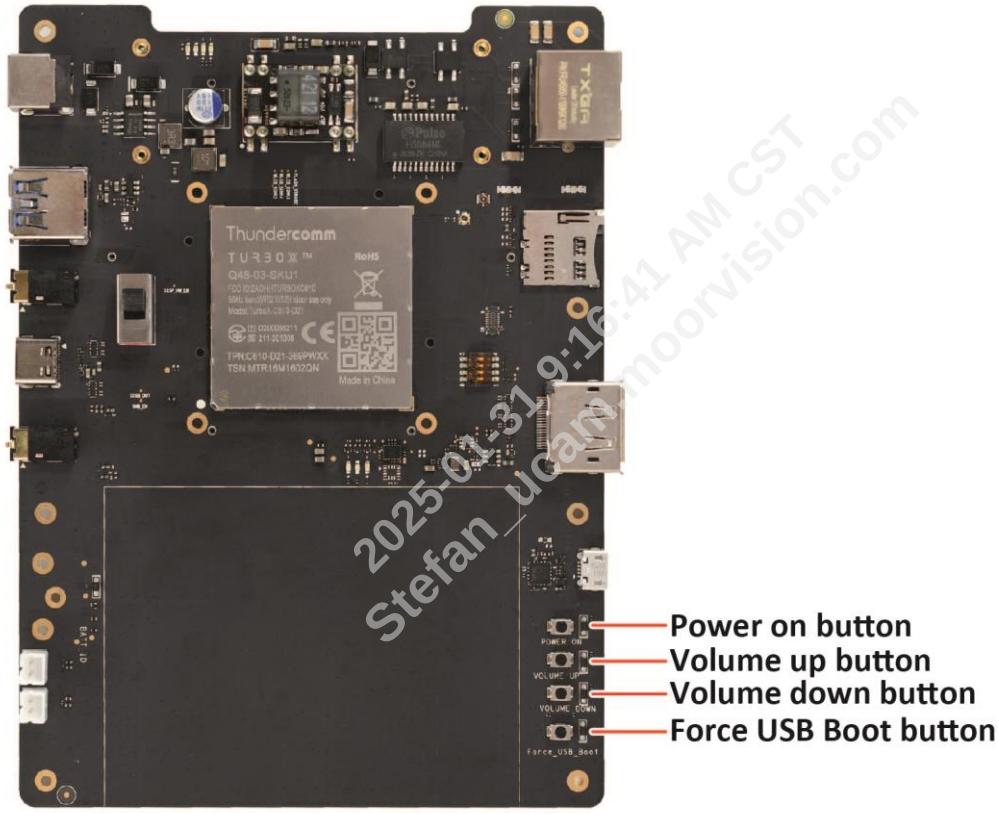


Figure 2-5. Button Layout

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Table 2-1. Button description

Button	Device node
Power on	/dev/input/event0
Volume up	/dev/input/event2
Volume down	/dev/input/event0
Force USB Boot	-

➲ **NOTE**: **Force USB BOOT** is used to force the system to enter emergency download mode.

To Get interrupt event, run the following commands:

```
$ adb shell
# cat dev/input/event0
# cat dev/input/event2
```

2.9. Wi-Fi/BT

2.9.1. Set up Wi-Fi STA mode

Step 1.Connect the antenna to the board. Refer to the figure below to locate antenna slot.

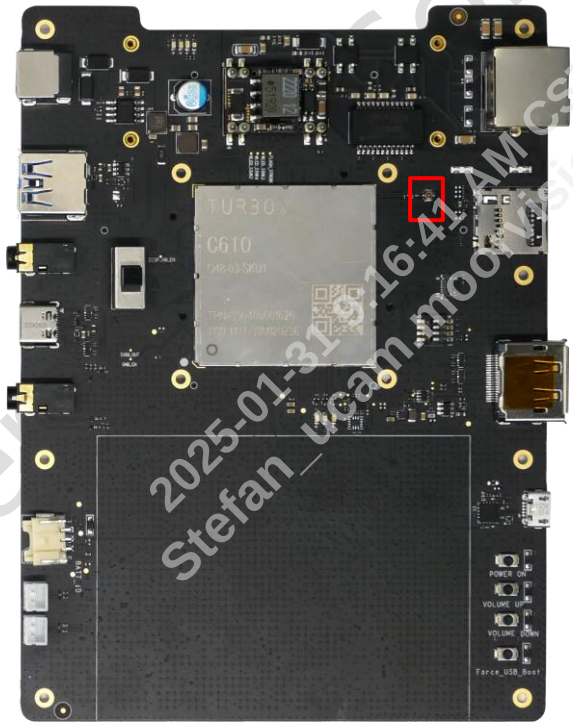


Figure 2-6.Antenna Slot

Step 2.Swipe downwards shortly from the top edge of your screen to enter **Quick Settings** bar.

Step 3.Tap the Wi-Fi icon to turn on Wi-Fi connection function.

Step 4.Holding the icon to enter network list.

Step 5.Tap the network you prefer to enter password (Networks that require password have a lock icon marked).

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Step 6.Confirm the connection with an IP obtained.

```
# ifconfig wlan0
wlan0     Link encap:Ethernet  HWaddr 8a:89:2b:ff:99:04  Driver icnss
          inet addr:192.168.1.101  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: fe80::c8:e1cf:5388:8125/64 Scope: Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:23 errors:0 dropped:0 overruns:0 frame:0
          TX packets:37 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:3000
          RX bytes:3418 TX bytes:4178
```

Step 7.Perform a ping test.

```
# ping www.baidu.com
PING www.baidu.com (23.59.93.51): 56 data bytes
64 bytes from 23.59.93.51: seq=0 ttl=48 time=171.722 ms
64 bytes from 23.59.93.51: seq=1 ttl=48 time=163.070 ms
64 bytes from 23.59.93.51: seq=2 ttl=48 time=244.932 ms
64 ytes from 23.59.93.51: seq=3 ttl=48 time=166.135 ms
```

2.9.2. Hotspot setup

Step 1.Swipe downwards shortly from the top edge of your screen to enter **Quick Settings** bar.

Step 2.Tap **Hotspot** icon.

➲ **NOTE**:

- If you cannot find the **Hotspot** icon, tap **Edit** at the bottom-left of your screen and drag **Hotspot** icon into your **Quick Settings** bar.
- You can change the AP name and password by tapping **Setting** > **Network and Internet** > **Hotspot & Tethering** > **Wi-Fi Hotspot**

Step 3.Use your phone to connect the Wi-Fi you have set up.

2.9.3. Connect to a Bluetooth-enabled device

Step 1.Swipe downwards shortly from the top edge of your screen to enter **Quick Settings** bar.

Step 2.Tap Bluetooth icon to turn on Bluetooth connection function.

Step 3.holding the icon to enter Bluetooth setting interface.

Step 4.Tap **Pair new device** to search the Bluetooth-enabled device around.

➲ **NOTE**: If you cannot find **Pair new device** option, check the Bluetooth list under **Available devices**.

Step 5.Tap the Bluetooth name to pair the Bluetooth-enabled device with your device.

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2.10. Camera module

Step 1.Refer to the figure below to install camera modules to the board.

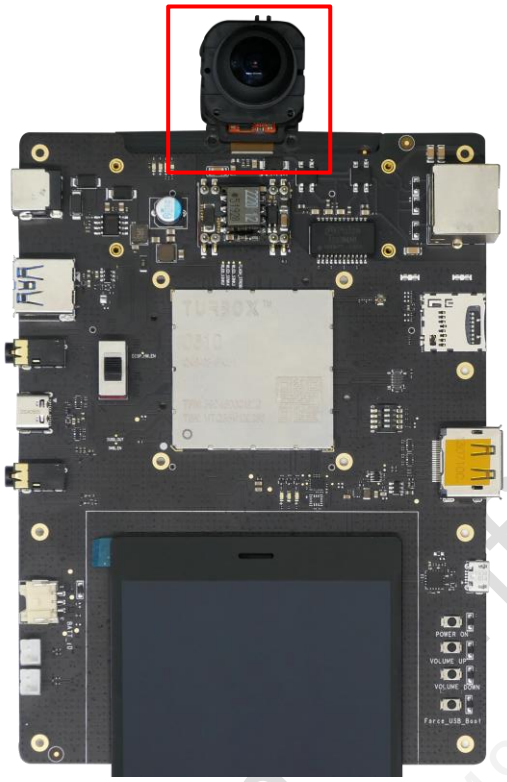


Figure 2-7.Camera Module Installation

Step 2.Set the switch for Sony IMX577 Camera Module Kit to the right position.

Step 3.Boot up your system, then you can open the app **Snapdragon Camera** to take picture or record video.

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2.11. Speaker

Step 1.Refer to the figure below to connect speakers to the board.

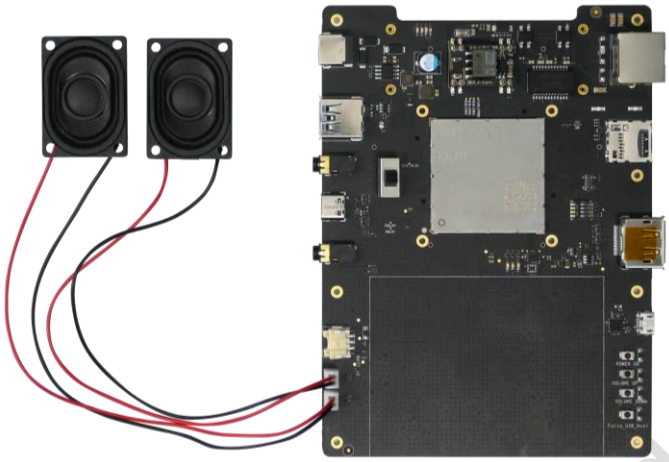


Figure 2-8.Speaker Installation

Step 2.Push an mp3 file to your device.

```
$ adb push test.mp3 /storage/emulated/0/Music
```

Step 3.If the **Music Player** fails to play the .mp3 file in **Music Player**, find the file at */storage/emulated/0/Music* and play the .mp3 file again.

2.12. POE (Power on Ethernet) connector

Step 1.Connect POE adapter to the board via POE connector.

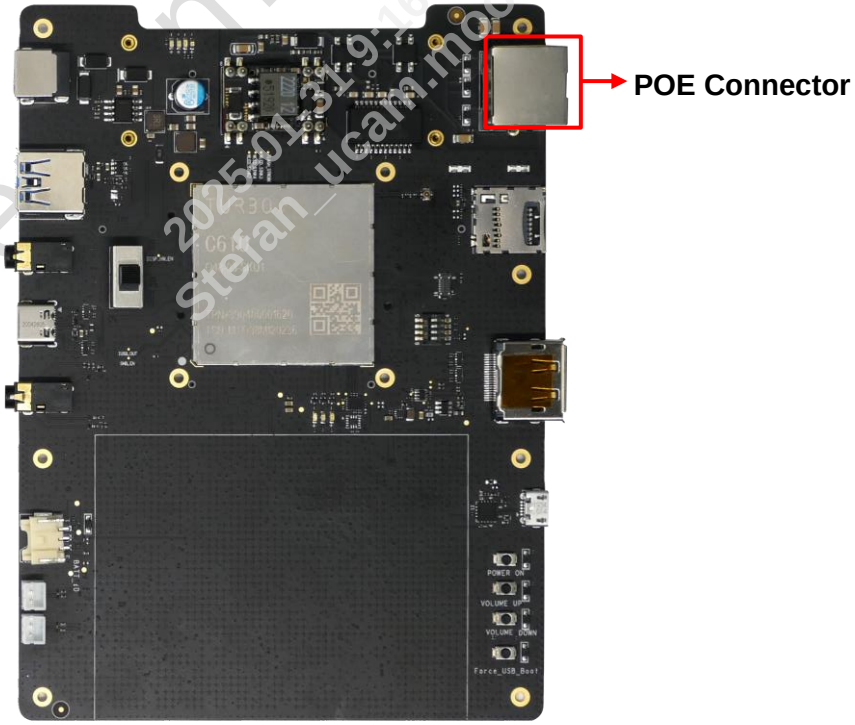


Figure 2-9.POE Connector Location

Step 2.Connect debug UART, press the power key to boot up system in normal mode.

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2.13. Accelerometer/gyroscope sensor

Step 1.Boot up the device system.

Step 2.Open the ***QsensorTest*** app to check Accel & Gyro sensor data. Refer to the figure below.

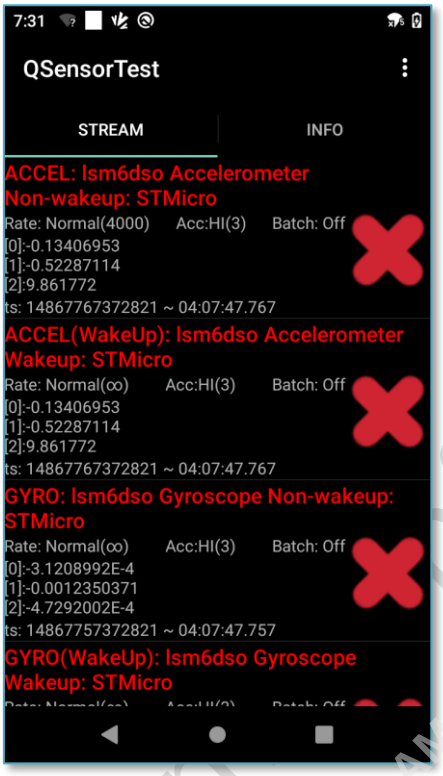


Figure 2-10. QsensorTest Screen Capture

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2.14. HDMI IN

Refer to Figure 2-11 for the following IMX415 camera board, which integrates the camera connector and HDMI-IN connector.

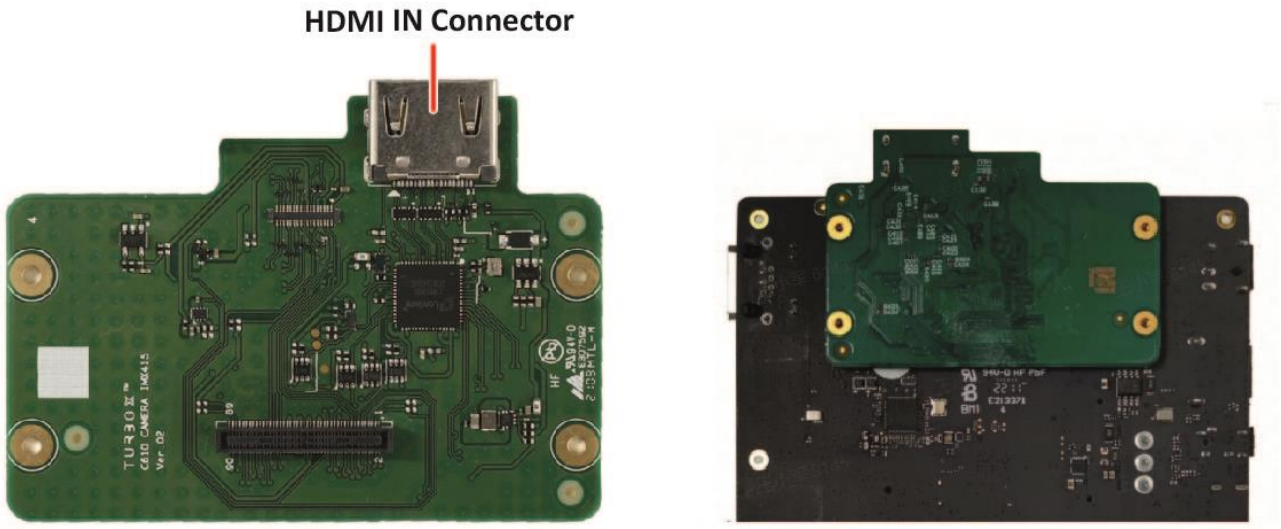


Figure 2-11. IMX415 Camera board

Step 1.Connect the host to the IMX415 camera board through the **HDMI IN connector** via an HDMI cable.

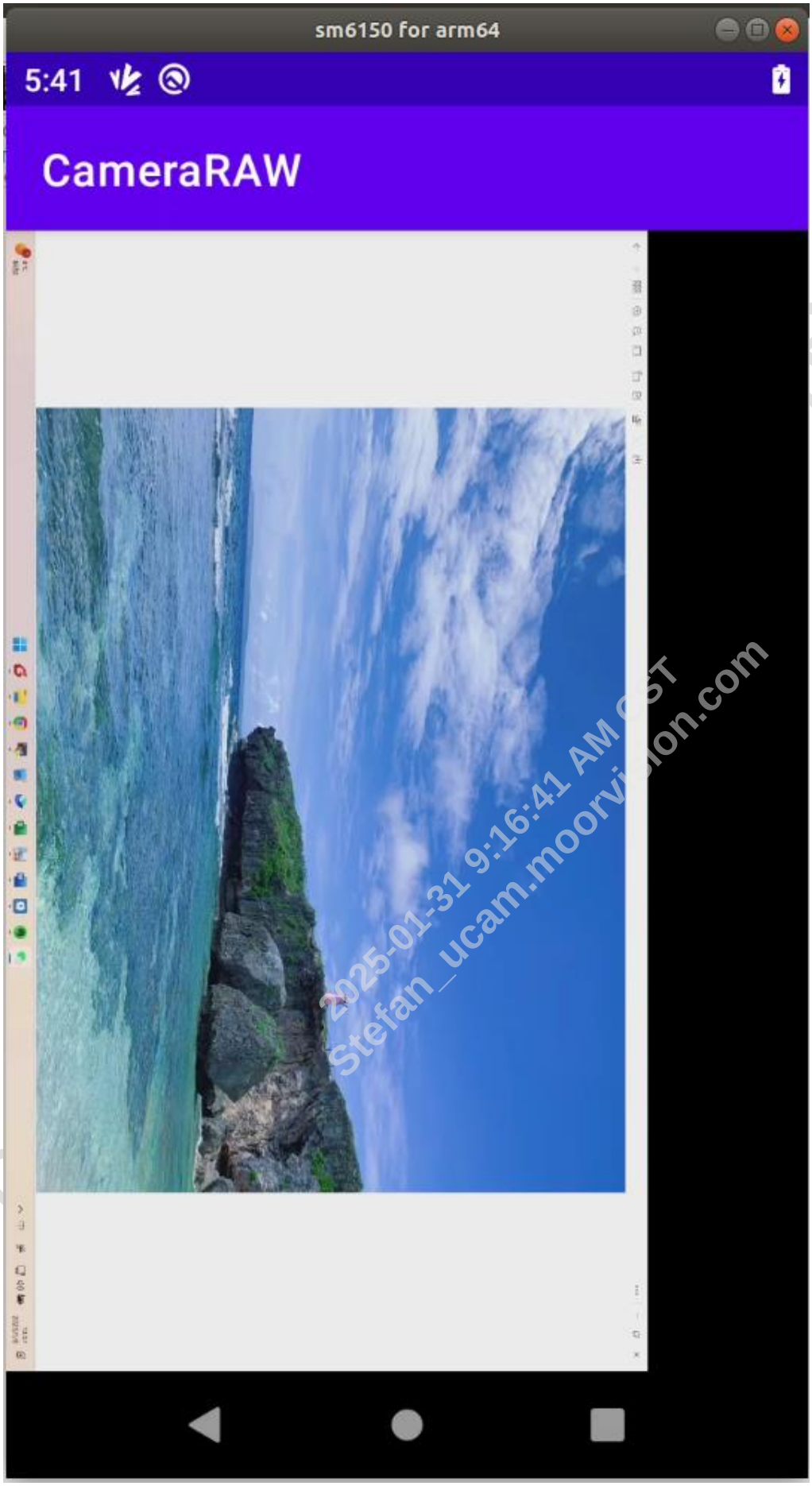
Then the host will prompt you to connect an external display.

Step 2.Run the CameraRAW application.

➲ **NOTE:**

- Before disconnecting the HDMI cable or before the system hibernates, you need to close the application first, otherwise the application cannot obtain the image correctly.
- This feature is currently available on LA1.0 only and will soon be supported by LA2.0.

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